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QUICKLY APPROXIMATING A LEARNING ALGORITHM WHILE PRESERVING ITS STATISTICAL RATE

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Abstract here.

1. Introduction. Scientific investigation, in many ways, concerns itself with prediction. Consequently, the task falls upon statisticians to construct effective predictive models. One might be interested in the outputs of such models for their own sake: to understand, for instance, whether an individual might respond to some treatment or policy. Or, these predictions might be useful as an input to some downstream task: for example, in summarizing how the *average* person might respond to a treatment.

A model is typically fit using some statistical learning algorithm. Such an algorithm is said to be an "estimator" of the function that the model represents. A good estimator should possess strengths in two areas: *theoretical* and *computational*. An estimator may be said to be theoretically robust if it has been mathematically proven to possess a fast (asymptotic) convergence rate—meaning that converges to the true, underlying target function quickly as sample size grows. An estimator is computationally efficient if it has a low time complexity, meaning that the amount of time that the algorithm takes to run does not grow too quickly in the sample size.

Oftentimes, these strengths do not overlap: an estimator that excels in terms of one desideratum will not perform well in the other. As an example, consider splines (or other models that minimize some loss over a basis). It has been long known that when the underlying data are multidimensional, tensor product smoothing splines can converge at fast rates to the true underlying function in broad function classes (Stone, 1994). But, in high-dimensions, tensor product splines that use regularization to select knot points generally have time complexity $O(n^2 \cdot 2^D)$: exponential in dimensionality D and quadratic in sample size n. Even when using the "kernel trick" to avoid dependence on the dimension of the data (Kimeldorf and Wahba, 1971a), the time complexity for such tensor-product splines will be $O(n^3)$. This limits their computational tractability in the analysis of data as collected in modern studies (e.g., electronic health records or -omics data).

On the other hand, models like random forests (Breiman, 2001) or factorization machines (Rendle, 2010) fit a regression over a smaller, more computationally-convenient basis. By relying on heuristics such as organizing the function into a series of trees or focusing on lower-order interactions, these algorithms provide much more practical fitting strategies. This practicality is evidenced by their widespread use and successful adoption across numerous applied science contexts.

Unfortunately, such basis selection heuristics severely complicate theoretical analysis. For example, in the context of tree-based learners, with many different possible function classes and heuristics for building trees, it is challenging to connect theoretical results to the types of algorithms typically deployed in practice. Mathematical efforts have

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typically focused on establishing dimension-dependent rates in the context of Hölder classes (Biau, Devroye and Lugosi, 2008; Gao, Xu and Zhou, 2022; Wager and Athey, 2018; Ročková and van der Pas, 2020), with some exceptions for estimation of specific restricted types of functions (Syrgkanis and Zampetakis, 2020), but the breadth of literature serves as a testament to the complexity of marrying theoretical results with practical and computationally performant algorithmic solutions.

Currently, many efforts begin by first considering a computationally convenient, well-performing prediction approach; then, they set out to prove its theoretical properties (which we term "working upwards"). In this work, we argue that it is often more practical to construct a suitable estimator by starting with an algorithm family that has already well-established theoretical properties (e.g., fast convergence rate) and constructing an approximation guaranteed to preserve those desirable properties (that is, by "working downwards"). Consequently, we take as our goal the development of a general framework, applicable to a wide array of algorithms, for constructing rate-preserving approximations of estimators of target functions. Our approach demonstrates how a practitioner can select a statistical learning algorithm and construct a computationally-efficient approximation that preserves the convergence rate of the original estimator.

As a case study, we apply this method to the Highly Adaptive Lasso (HAL), a type of lasso-penalized tensor product spline model that achieves fast rates in a particular function class. We develop RandomHAL, a random basis approximation of the original HAL algorithm that preserves its fast rate while running in a more efficient $O(dn \log n)$ time complexity, matching that of typical tree-based algorithms. As it turns out, many existing practical tools for statistical learning, including decision trees and the multivariate adaptive regression splines (MARS) (Friedman, 1991) algorithm, can be conceptualized as specific sampling schemes within the RandomHAL algorithm. This provides theoretical insight into why such methods perform so well in practice.

Contributions. To faciliate the analysis of a rate-preserving approximations, we describe an "approximation process": a sequence of loss minimizers over a growing subset of the parameters of a full model. This supports a few mathematical contributions. First, we describe Markov- and Chernoff-style probabilistic error bounds for the loss-based dissimilarity of such a sequence of estimators that relies on minimal assumptions about the nature of the loss. Second, we describe finite-sample expected error bounds for optimizers over a random subset of parameters chosen uniformly with and without replacement. Third, in the case of a convex loss function, we present a theorem when certain subsets of parameters can be safely excluded from the sampling process without affecting the asymptotic rate of the estimator. In the case of a model linear over a particular basis, this leads to useful heuristics and an efficient sampling scheme that reduces the time complexity for fitting such a model to $O(n \log n)$.

Paper organization. After a background literature review and notational definitions (Section 2), we organize this manuscript as follows. Section 3 describes the notions of early-stopping and rate-preserving approximation processes, including our Markov- and Chernoff-style probabilistic error bounds. Section 4 describes finite-sample error bounds over subsets of parameters: both uniformly sampled subsets and subsets where specific unfavorable parameters may be dropped. In Section 5, we apply these principles to practical example: constructing a rate-preserving approximation of a lasso-regularized tensor product spline (the "highly adaptive lasso"). We call this approximation "RandomHAL". In conjunction, Section 6 describes how a random approximation can be constructed such that its time complexity matches the $O(n \log n)$ of common tree-based models. Finally, we present numerical simulations (Section 7) to demonstrate the efficacy of our method on tensor product splines, and conclude with a discussion (Section 8).

2. Background and notation. Consider observing a random collection of identically-distributed data $O = [O_i]_{i=1}^n$. Formally, our goal will be to estimate (learn) a function $f^\star$ that minimizes an expected loss L within a function class $\mathscr{F}$; that is,

$$f^\star = \arg\min_{f \in \mathscr{F}} \mathsf{E}[L(f; O)]$$

Typically, one will proceed by constructing an estimator $\hat{f}$ to minimize the empirical loss:

$$\hat{f} = \arg\min_{f \in \mathscr{F}} L(f; O)$$

For instance, suppose $O_i = (X_i, Y_i)$. As one example, one might want to learn the conditional expectation function

$$\mathsf{E}(Y_i \mid X_i) = \arg\min_{f \in \mathscr{F}} \mathsf{E}[(Y_i - f(X_i))^2]$$

by minimizing the empirical loss $n^{-1} \sum_{i=1}^n (Y_i - f(X_i))^2$. As a second example, if one sought to learn the conditional probability

$$\mathbb{P}(Y_i \mid X_i) = \arg\min_{f \in \mathscr{F}} \mathsf{E}[Y_i \log(f(X_i)) + (1 - Y_i) \log(1 - f(X_i))]$$

they could minimize $n^{-1} \sum_{i=1}^n Y_i \log(f(X_i)) + (1 - Y_i) \log(1 - f(X_i))$, the negative log-likelihood (or cross-entropy) loss.

Since these optimization problems typically must be solved on a computer, it is common to represent functions $f \in \mathscr{F}$ by some finite vector of parameters $\beta \in \mathbb{R}^d$, so that $\mathscr{F} = \{f_\beta : \beta \in R \subset \mathbb{R}^d\}$. Overloading notation, the loss can be represented by $L(\beta; O)$ instead; consequently, we are interested in estimating $f^\star = f_{\beta^\star}$ by solving

$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^d} L(\beta; O) \ ,$$

The quality of $f_{\hat{\beta}}$ as an estimator—or, conversely, the difficulty of estimating a function f in $\mathscr{F}$—is often measured by the rate r_n at which $f_{\hat{\beta}}$ converges to $f^\star$ in loss-based dissimilarity as the sample size n grows—that is,

$$\mathsf{E}(L(\hat{\beta}) - L(\beta^\star)) = \|\hat{f} - f^\star\|^2 = o_{\mathsf{P}}(r_n^2) \ .$$

This rate r_n is controlled by the complexity of the function class $\mathscr{F}$, creating a tradeoff—the more flexible and complicated f (and consequently, the richer $\mathscr{F}$) is allowed to be, the slower the best estimation possible rate will be. For parametric problems in which d is fixed, loss-based estimation schemes typically achieve a fast rate, $r_n = n^{-1/2}$.

But in more flexible function classes, the story can change. For example, if $f \in \mathscr{F}$ is only assumed to have a certain degree α of smoothness (i.e., an α-Hölder condition), the best possible rates typically will be $r_n = n^{-\alpha/(\alpha+d)}$ (Yang and Barron, 1999). Occasionally termed the "asymptotic curse of dimensionality," these best achievable rates may be slow in settings with more than only a handful of covariates.

Function Class	Minimax Optimal Rate	Example
Parametric	$n^{-1/2}$	Linear Regression
Cadlag $\|\beta\|_1 < \infty$	$n^{-1/3}(\log n)^{3(d-1)/2}$	Highly Adaptive Lasso
α-Hölder	$n^{-\alpha/(\alpha+d)}$	Random Forest

TABLE 1
Rates for function classes used to analyze common regression models.

To obtain both flexibility and a fast asymptotic convergence rate, one can employ integrability assumptions. For example, suppose $\mathscr{F}$ consists of functions absolutely continuous over each section; that is,

$$f(x) = f(a) + \int_a^x f'(t)dt = f(a) + \sum_{S \subset \{1,\ldots,d\}} \int_{a_S}^{x_s} f'_S(t)dt \ . \tag{1}$$

This is equivalent to $\mathscr{F}$ being the set of càdlàg functions with bounded sectional variation, a construction used to develop the highly adaptive lasso (HAL) estimator (van der Laan, 2015; Benkeser and van der Laan, 2016; van der Laan, 2017; Qiu, Luedtke and Carone, 2021; van der Laan, 2023). If f has k derivatives, it can be represented as the lasso-constrained tensor product spline:

$$f(x) = \sum_{i=1}^{n} \sum_{S \subset \{1,\ldots,d\}} \prod_{j \in S} (x_j - X_{ij})^k \mathbb{I}(x_j \leq X_{ij}) \beta_{i,S} \text{ with } \|\beta\|_1 \leq \infty \ , \tag{2}$$

which, after the basis has been constructed, can be fit using standard routines for lasso regression. Such an estimator achieves the minimax rate for this function class, $n^{-1/3} \log(n)^{3(d-1)/2}$ (Bibaut and van der Laan, 2019; Fang, Guntuboyina and Sen, 2021). Since this rate depends minimally on the dimension d of the data, it often strikes a useful balance between flexibility and convergence rate. Using instead an assumption of bounded mixed derivatives, which imposes an l_2 penalty on the tensor product spline, yields the highly adaptive ridge (HAR) (Schuler, Hagemeister and van der Laan, 2024) estimator with a similar convergence rate. Such results can be obtained in Sobolev spaces as well, which also yield similar rates with a basis of cosine functions (Zhang and Simon, 2023).

An important aside: in the field of semiparametric statistics, beyond being able to achieve low risk with fewer samples, we note that achieving a fast r_n is often imperative when using $f_{\hat{\beta}}$ in some downstream statistical inference problem. For example, a common class of asymptotically efficient estimators in semi-parametric estimation, especially causal inference and missing data problems, involves estimating equations of the form.

$$\frac{1}{n} \sum_{i=1}^{n} \phi(X_i; \eta, \psi) = 0 \ , \tag{3}$$

where ϕ is called an "influence function", η a collection of two or more nuisance functions $f_1, f_2, \ldots$, and ψ is the true parameter to be estimated. See, for instance, Koshevnik and Levit (1977); Pfanzagl and Wefelmeyer (1985); Klaassen (1987); van der Laan and Robins (2003); van der Laan and Rubin (2006); van der Laan and Rose (2011); Chernozhukov et al. (2018). These aforementioned works set up such estimating equations so that they will converge asymptotically to a normal distribution centered around ψ provided that the functions in η can be estimated at a fast enough rate. Specifically, ϕ is often "rate doubly-robust" meaning that for $\eta = (f_1, f_2)$, asymptotic normality is achieved if the product of both functions' rates satisfies $\|\hat{f}_1 - f_1^\star\| \|\hat{f}_2 - f_2^\star\| = o_{\mathsf{P}}(n^{-1/2})$.

Unfortunately, finding the best solution to a statistical problem by optimizing over the entire function class can be computationally prohibitive. Oftentimes, the function class $\mathscr{F}$ is so "large" that it requires too many parameters to represent f; this occurs, for instance, even when the underlying set of covariates X in a dataset $O = (X, Y)$ is of moderate size. Tensor product splines (Stone, 1994) form the prototypical example on which the illustration of Section 5 will focus: when X is of dimension D, the tensor product creates a basis where $d = O(2^D)$. As a tensor product spline estimator, fitting

HAL has a time complexity of $O(n^2 \cdot 2^D)$. Similar exponential scaling issues occur in multidimensional wavelets (Kugarajah and Zhang, 1995; Selesnick, Baraniuk and Kingsbury, 2005) and other tensor product models (Lin, 2000), as well as the space of all possible decision trees (Breiman et al., 2017).

In some cases, the "kernel trick" can be used to reduce a high-dimensional basis to a kernel matrix (Mercer, 1909), from which the function admits representation (Kimeldorf and Wahba, 1971b) and dimension-independent computation (Boser, Guyon and Vapnik, 1992). This technique encompasses kernel ridge regression, support vector machines (Cortes and Vapnik, 1995), and Gaussian process regression (Williams and Rasmussen, 1995). However, it is only available when $\mathscr{F}$ is reproducing kernel Hilbert space (RKHS), which limits possible regularization schemes to norms induced by an inner product (i.e., l_2 norms). Additionally, optimization over an $n \times n$ kernel matrix typically requires $O(n^3)$ computation, which is not computationally feasible in settings with all but the smallest sample sizes.

As a consequence, computationally-efficient statistical learning algorithms typically rely on some approximation heuristics for fast training. Commonly used tree-based algorithms like decision trees (Breiman et al., 2017), random forests (Breiman, 2001), XGBoost (Chen et al., 2025), and their Bayesian variants (Chipman, George and McCulloch, 2010), as well as spline-based algorithms like multivariate adaptive regression splines (MARS) (Friedman, 1991) rely on the heuristic of *greedily* sampling a subset of basis functions that admit fast computation.

But such algorithms possess another problem: analysis of their theoretical convergence rates often must be performed piecemeal across different estimators and function classes. For instance, convergence rates derived for the random forests algorithm (Breiman, 2001; Biau, Devroye and Lugosi, 2008; Biau, 2012; Wager and Athey, 2018; Athey, Tibshirani and Wager, 2019; Syrgkanis and Zampetakis, 2020; Peng, Coleman and Mentch, 2022) each focus on specific versions of the random forest algorithm. Often, it may not be clear whether a particular implementation of an algorithm obtains a necessary rate—especially when, for example, an α-Hölder smoothness rate may not suffice to ensure the $o_{\mathsf{P}}(n^{-1/4})$ rate needed in common semi-parametric estimating equations.

To avoid the labor of deriving rates on case-by-case basis, an alternative approach might instead inform as to how to modify an existing algorithm that already attains a desired convergence rate to achieve better computational efficiency—without altering the rate. A number of papers apply this idea to the HAL basis: Qiu (2023) propose a method to reduce the memory burden, though computational speed remains slow; Schuler, Hagemeister and van der Laan (2024) kernelizes the tensor spline basis and Wang et al. (2026) performs a principal components regression the basis, but these still require $O(n^3)$ time complexity; and finally, Schuler, Li and van der Laan (2023) show that fitting the lasso over a series of boosted trees achieves the same rate as HAL, but do not analyze how many trees may be needed to accomplish this.

We adopt and investigate this alternative perspective. Specifically, a simpler strategy often might be to construct a *rate-preserving approximation* that solves the same optimization problem, but over a small, computationally-tractable subset of all d parameters. The idea has roots in the context of rate-preserving "early stopping estimators" described in the context of boosting by Bühlmann and Yu (2003) and Raskutti, Wainwright and Yu (2014). Such works explore the notion of a functional gradient descent, run over a finite number of iterations, in the context of linear basis functions and reproducing kernel Hilbert spaces, respectively. However, rather than relying on a direct asymptotic analysis, we assume our algorithm approximates a full model already known to have a fast rate; hence, our work falls more closely in line to the types of rate-preserving approximations introduced by Schuler, Li and van der Laan (2023). This idea also relates to the work on random features by Rahimi and Recht (2007, 2008), who showed that a linear model fit over a relatively small random basis can be quite effective at approximating certain classes of functions. *** Nima: stopped here—5th january

3. Early-stopping approximation process. Suppose our function estimator $f_{\hat{\beta}}$ theoretically converges quickly to $f^\star$, but is too computationally intensive to calculate in practice. In this case, we can consider an "early stopping" estimator $f_{\hat{\beta}}^{(m)}$ that optimizes only m parameters of $f_{\hat{\beta}}$. By the triangle inequality, the error of this estimator is bounded by

$$\|f^\star - f_{\hat{\beta}}^{(m)}\| \leq \underbrace{\|f - f_{\hat{\beta}}\|}_{\text{estimation error}} + \underbrace{\|f_{\hat{\beta}} - f_{\hat{\beta}}^{(m)}\|}_{\text{approximation error}} = o_P(\min\{r_n, r_m\}) \ , \tag{4}$$

where we refer to the rates as follows:

- r_n is the *estimation rate*: $\|f^\star - f_{\hat{\beta}}\| = o_P(r_n)$
- r_m is the *approximation rate*: $\|f_{\hat{\beta}} - f_{\hat{\beta}}^{(m)}\| = o_P(r_m)$

For our random approximation $f_{\hat{\beta}}^{(m)}$ to achieve the same asymptotic rate as $f_{\hat{\beta}}$, m must to be set such that the approximation error converges at the same rate or faster than the estimation error. Therefore, our goal in this section will be to bound the error of a random approximation. In general, the norms above are equivalent to the loss-based dissimilarity; that is,

$$\|f_{\hat{\beta}} - f_{\hat{\beta}}^{(m)}\|^2 = L(\hat{\beta}^{(m)}) - L(\hat{\beta}) \tag{5}$$

To study how the excess loss of our approximation changes as one includes an increasing number of parameters into the approximation, we can conceptualize adding parameters as a discrete random process. For notational brevity, denote the excess loss after a growing series of m parameter updates as

$$X_m = L(\hat{\beta}^{(m)}) - L(\hat{\beta}) \ .$$

We term this random object an *approximation process*, as it describes the behavior of a growing random approximation. Algebraically, one can express the update scheme as the supermartingale

$$X_{k+1} = X_k + D_k = X_0 + \sum_{j=1}^{k+1} D_j \ ,$$

where $D_k = L(\beta^{(k+1)}) - L(\beta^{(k)})$ represents the nonpositive change in loss at each step.

If the approximation process becomes small, one can be assured that the loss of the early-stopping estimator will be close to the full model. When does this occur? Let $\mathcal{F}_k$ denote a filtration containing the random parameter updating scheme at each step of the process. Some basic finite-sample bounds can be established by considering bounds on the expected difference $E[D_k \mid \mathcal{F}_{k-1}]$. Suppose

$$\mathsf{E}[D_k \mid \mathcal{F}_{k-1}] \leq \delta_k X_{k-1}$$

In this case, two probabilistic error bounds may be established: one based on Markov's inequality, and other based on the Chernoff bound.

THEOREM 1 (Large Approximation in Expectation). If X_m represents the excess loss after a sequence of growing parameter updates according to a filtration $\mathcal{F}_m$, X_0 is a fixed starting value, and $\mathsf{E}[D_k \mid \mathcal{F}_{k-1}] \leq \delta_k X_{k-1}$, then

$$\mathsf{E}[X_m] \leq \exp\Big(-\sum_{k=1}^{m} \delta_k\Big) X_0 \ .$$

By Markov's inequality, one could use Theorem 1 to establish the probabilistic bound

$$\mathbb{P}(X_m \geq \varepsilon) \leq \frac{\exp\Big(-\sum_{k=1}^{m} \delta_k\Big) X_0}{\varepsilon} \ .$$

However, further analysis can establish a more precise probabilistic error bound.

THEOREM 2 (Large Approximation Probabilistic Error Bound). If, in addition to the assumptions of Theorem 1, the update D_k is totally bounded such that

$$|D_k - \mathsf{E}(D_k \mid \mathcal{F}_{k-1})| \leq c_k \ ,$$

then it follows that

$$\mathbb{P}(X_m \geq \varepsilon) \leq \exp\Big(\frac{-(\varepsilon - \exp(\sum_{k+1}^{m} \delta_k)/X_0)^2}{2\sum_{k=1}^{m} c_k^2}\Big) \ .$$

Both theorems hinge on a simple intermediary lemma, which we present and prove.

LEMMA 1 (Product Bound). Given a series of $\delta_k \in [0, 1)$,

$$\prod_{k=1}^{m}(1-\delta_k) \leq \exp\Big(-\sum_{k=1}^{m} \delta_k\Big) \text{ and } \prod_{k=1}^{m}(1+\delta_k) \leq \exp\Big(\sum_{k=1}^{m} \delta_k\Big) \ .$$

PROOF OF LEMMA 1. Since $1 - \delta_k \geq 0$, we can apply the AM-GM inequality and the fact that, by Taylor expansion, $\exp(-x) \geq (1 - x/m)^m$ to see that

$$\prod_{k=1}^{m}(1-\delta_k) \leq \Big(\frac{1}{m}\sum_{k=1}^{m}(1-\delta_k)\Big)^m = \Big(1 - \frac{\sum_{k=1}^{m} \delta_k}{m}\Big)^m \leq \exp\Big(-\sum_{k=1}^{m} \delta_k\Big)$$

The same logic follows if $-$ is replaced with $+$. □

PROOF OF THEOREM 1. By the bound $\mathsf{E}[D_k \mid \mathcal{F}_{k-1}] \leq \delta_k X_{k-1}$ and the law of iterated expectations,

$$\mathsf{E}[X_{k+1}] = \mathsf{E}[X_k - \mathsf{E}[D_k \mid \mathcal{F}_{k-1}]] \leq (1-\delta_k)\mathsf{E}[X_k]$$

Applying this definition recursively followed by Lemma 1,

$$\mathsf{E}[X_m] \leq \prod_{k=1}^{m}(1-\delta_k) X_0 \leq X_0 \exp\Big(-\sum_{k=1}^{m} \delta_k\Big) \ .$$

Plugging in $\mathsf{E}[X_m]$ to Markov's inequality yields the result. □

Proof of Theorem 2. Our goal will be to bound moment generating functions of X_m and D_k in order to apply the Chernoff bound. First, we bound the MGF of $D_k - \mathsf{E}[D_k \mid \mathcal{F}_{k-1}]$ conditional on the filtration. If $|D_k - \mathsf{E}[D_k \mid \mathcal{F}_{k-1}]| \le c_k$, then it must be c_k-subgaussian Wainwright (2019), meaning that

$$\mathsf{E}\Big[\exp(\lambda(D_k - \mathsf{E}[D_k \mid \mathcal{F}_{k-1}])) \mid \mathcal{F}_{k-1}\Big] \le \exp\left(\lambda^2 c_k^2/2\right) .$$

This will allow use to bound the entire MGF of X_m. Adding and subtracting the expectation conditional on the filtration $\mathsf{E}[D_k \mid \mathcal{F}_{k-1}]$ and applying the law of iterated expectations,

$$\begin{aligned}
\mathsf{E}[\exp(\lambda X_{k+1})] &= \mathsf{E}\Big[\exp\left(\lambda(X_{k+1} - \mathsf{E}[D_k \mid \mathcal{F}_{k-1}] + \mathsf{E}[D_k \mid \mathcal{F}_{k-1}])\right)\Big] \\
&\le \mathsf{E}\Big[\underbrace{\exp\left(\lambda(X_k + \mathsf{E}[D_k \mid \mathcal{F}_{k-1}])\right)}_{\le \exp(\lambda(1+\delta_k)X_k)} \underbrace{\mathsf{E}\Big[\exp(-\lambda(D_k - \mathsf{E}[D_k \mid \mathcal{F}_{k-1}])) \mid \mathcal{F}_{k-1}\Big]}_{\le \exp(\lambda^2 c_k^2)}\Big] \\
&\le \exp(\lambda^2 c_k^2)\mathsf{E}[\exp(\lambda(1+\delta_k)X_k)] \ .
\end{aligned}$$

Therefore, by recursion and Lemma 1,

$$\begin{aligned}
\mathsf{E}[\exp(\lambda X_m)] &\le \exp\left(\lambda \prod_{k=1}^{m}(1+\delta_k)X_0 + \lambda^2 \sum_{k=1}^{m} c_k^2\right) \\
&\le \exp\left(\lambda \exp\left(\sum_{k=1} \delta_k\right) X_0 + \lambda^2 \sum_{k=1}^{m} c_k^2\right)
\end{aligned}$$

Applying the Chernoff bound,

$$\mathbb{P}(X_m \ge \varepsilon) \le \inf_{\lambda>0} \exp\left(\lambda \exp\left(\sum_{k=1} \delta_k\right) X_0 + \lambda^2 \sum_{k=1}^{m} c_k^2 - \lambda\varepsilon\right) .$$

Minimizing the right-hand side yields the desired result. □

As an investigator, our interest will be in constructing an early-stopping estimator such that the approximation process converges asymptotically to zero. Let us examine how to design such an early-stopping scheme.

4. Asymptotic behavior of an approximation process. To develop a fast early-stopping scheme that preserves the rate of the original estimator, our goal will be to construct an approximation process that vanishes asymptotically in n. To accomplish this, one might consider two strategies: (1) choosing the number of updated parameters m to be sufficiently proportional to n, and (2) leaving out parameters that, when sampled, yield a loss reduction D_k that is asymptotically negligible. Let us start by considering strategy (1).

4.1. *Asymptotics in m.* A sufficient choice of m will depend on whether parameters may be sampled with or without replacement. In the case that it is possible to sample parameters *without* replacement, an obvious yet imperative remark follows.

Remark 1 (Sampling without replacement). If the full model $f_{\hat\beta}$ contains $O(d_n)$ parameters and converges in loss-based dissimilarity at rate $o_p(1/r_n^2)$, then any random approximation $f_{\hat\beta}^{(m)}$ optimizing over $m \gtrsim d_n$ parameters will also converge at rate $o_p(1/r_n^2)$.

This is because choosing $m \gtrsim d_n$ guarantees that, asymptotically, $\hat{f}_{\hat{\beta}}^{(m)}$ will eventually exhaust every parameter in the full model, driving the approximation error to zero. Though seemingly trivial, this tells us that, in fact, many broad classes of sampling schemes preserve asymptotic rates. For example, the rate of a 0-order tensor product spline can be preserved by solving the same optimization problem over a growing subset of the basis. Schuler, Li and van der Laan (2023) exploit this idea to show that, since random trees asymptotically exhaust the 0-order tensor product spline basis, boosted decision trees can achieve the desirable $o_P(n^{-1/4})$ rate of HAL. More generally, this rate is achieved asymptotically by any estimator that solves a regularized regression problem over a $m \gtrsim n$ set of step functions chosen uniformly without replacement.

How fast does the error decay when sampling without replacement? We can derive a finite-sample error update under uniform sampling as follows.

THEOREM 3 (Expectation bound for sampling without replacement). Suppose an approximation $f_{\hat{\beta}}^{(m)}$ that samples m parameters uniformly at random without replacement from $\beta_1, \ldots, \beta_d$ to optimize. Then,

$$\mathsf{E}[X_m] \leq \left(1 - \frac{m}{d}\right) X_0 \ .$$

PROOF. Let S_k denote a set of k parameters sampled without replacement such that $S_k \subset \{1, \ldots, d\}$ and $|S_k| = k$, and denote by $L(S)$ the optimal loss over the subset of parameters S. Clearly,

$$\begin{aligned} \mathsf{E}[L(\beta^{(k)}) - L(\beta^{(k+1)}) \mid \beta^{(k)}] &= \mathsf{E}[L(S_k) - L(S_{k+1}) \mid S_k] \\ &= \frac{1}{d-k} \sum_{j \in S_d - S_k} \Big(L(S_k) - L(S_k - \{j\}) \Big) \ , \end{aligned} \tag{6}$$

since if k parameters have already been sampled, and one more is added at $k+1$, only $d-k$ parameters will be available to sample at the next step.

Let $j_1, \ldots, j_d$ be an arbitrary permutation of the parameter indices, and $W_i = \{j_1, \ldots, j_i\}$ be an arbitrary "walk" through the parameters in the order given by the permutation. Then, the difference between the optimal loss at S_k and the optimum at the full set of parameters S_d can be bounded using a telescoping series:

$$\begin{aligned} L(S_k) - L(S_d) &= \sum_{i=0}^{d-k-1} L(S_k \cup W_i) - L(S_k \cup W_{i+1}) \\ &\leq \sum_{i=0}^{d-k-1} L(S_k) - L(S_k \cup W_{i+1}) \\ &\leq \sum_{j \in S_d - S_k} L(S_k) - L(S_k \cup \{j\}) \end{aligned} \tag{7}$$

The second step holds because $L(S_k \cup W_i) \leq L(S_k)$ (optimizing a new parameter cannot worsen the loss) and the third step because $L(S_k \cup W_{i+1}) \leq L(S_k \cup \{j\})$ for all j (the change in loss from updating parameter j can be no better than the change from updating j and every other parameter before it).

But, Line 7 is equivalent to Equation 6, scaled by $d-k$, meaning that

$$D_k \leq \frac{1}{d-k} \Big(L(S_k) - L(S_d) \Big) = \frac{1}{d-k} X_k \ .$$

Applying Theorem 1 with $\delta_k = 1/(d-k)$, the expected value will depend on the sum

$$\sum_{k=1}^{m} \frac{1}{d-k} = \sum_{k=1}^{d} \frac{1}{k} - \sum_{k=1}^{d-m} \frac{1}{k} \leq \log(d) - \log(d-m) = \log\left(\frac{d}{d-m}\right) ,$$

which follows from the harmonic series partial sum bound (Boas and Wrench, 1971)

$$\sum_{k=1}^{d} \frac{1}{k} \leq \log(d) + \gamma ,$$

where γ denotes the Euler-Mascheroni constant. Finally, plugging this into Theorem 1, we have that

$$\mathsf{E}[L(\beta^{(m)}) - L(\hat{\beta})] \leq \exp\Big(-\log\Big(\frac{d}{d-m}\Big)\Big) X_0 = \Big(1 - \frac{m}{d}\Big) X_0 ,$$

completing the proof. □

Theorem 3 tells us that, under uniform sampling without replacement, the approximation error will decay at most linearly before exhausting the entire basis. In some cases, however, the full set of parameters may be difficult to sample without replacement, perhaps because parameters may be difficult to index due to the size or complexity of the model. Consequently, we can consider sampling with replacement.

THEOREM 4 (Expectation bound for sampling with replacement). Suppose an approximation $f_{\hat{\beta}}^{(m)}$ that samples m parameters uniformly at random *with* replacement from $\beta_1, \ldots, \beta_d$ to optimize. Then,

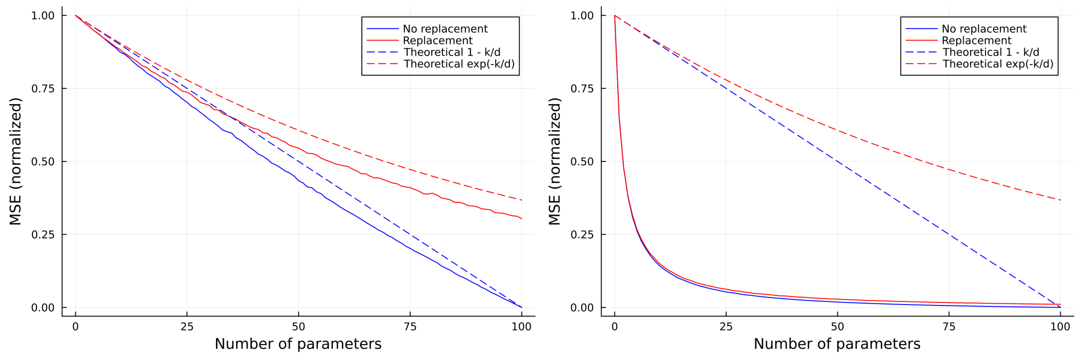
$$\mathsf{E}[X_m] \leq \exp(-m/d) X_0 .$$

PROOF. The proof follows almost identically to Theorem 3, replacing the mean weights $\delta_k = 1/(d-k)$ that arise from sampling without replacement with $\delta_k = 1/d$. □

Theorem 4 tells us that an extra $\log(n)$ terms must be sampled to overcome the possibility of sampling the same parameter multiple times under replacement.

Are these bounds useful? Figures 1a and 1b demonstrate the tightness of the theoretical bounds of Theorems 4 and 3 relative to simulated data for an OLS model in two synthetic data setups. When covariates were totally uncorrelated, the empirical average mean-squared error between a linear model using all covariates and a model using only a subset of the parameters matched very closely with the worst-case bound. Hence, reducing the approximation error to an arbitrarily small quantity requires sampling almost every covariate. However, in a setting where all covariates were moderately correlated with each other ($\rho = 1/9$), the theoretical bounds were extremely loose, and a small error could be obtained only sampling a relatively small number of parameters. In light of this result, the next section will explore an alternative strategy to constructing early-stopping approximations that whose error remains asymptotically negligible.

Even though they describe the tail behavior for an extremely broad class of estimators for a given choice of m, the bounds constructed using Theorem 3 and 4 are weak. In unfavorable cases, the error will not be bounded at a reasonable magnitude until one has sampled almost every parameter. Can one still obtain good performance even with much smaller m?



(a) Uncorrelated covariates (b) Correlated covariates ($\rho = 1/9$)

Fig 1: Average MSE between full model and uniformly-sampled early-stopping approximation for a linear model over 100 covariates with $n = 400$. Simulation run with 500 replicates, covariates drawn from a multivariate normal distribution, and outcome from a raised cosine distribution.

4.2. *Asymptotics after excluding certain D_k.* Instead of considering how an approximation process behaves solely in m, we can instead consider the asymptotic behavior of individual updates after sampling a certain parameter. If the update D_k conditional on sampling a certain parameter at step k is asymptotically negligible, then that parameter can be safely excluded from the model. Such a phenomenon can arise by enforcing relatively mild restrictions on the structure of the optimization problem.

THEOREM 5 (Subset sampling). Let $L(\beta) = \ell(\beta) + R(\beta)$ be some convex loss with subderivative ∇L whose solution enforces $C_1\|\beta\|_2 \leq R(\beta) \leq C_2$. Suppose at step k, the approximation process samples the jth parameter in β. Then,

$$|D_k| \leq \frac{2C_2}{C_1}\|\nabla_j L(\beta^{(k)})\|_2 \ .$$

PROOF OF THEOREM 5. Since the loss is convex,

$$D_k = L(\beta^{(k+1)}) - L(\beta^{(k)}) \geq \nabla L(\beta^{(k)})^\top(\beta^{(k+1)} - \beta^{(k)}) \ .$$

Since we only update the subset S_{k+1} of parameters at step $k+1$, the summands over parameters not yet sampled equal zero, implying

$$D_k \geq \sum_{j \in S_k} \nabla_j L(\beta_j^{(k)})(\beta_j^{(k+1)} - \beta_j^{(k)}) \ .$$

Since this value must always be negative, we can bound it using the Cauchy-Schwarz inequality.

$$\begin{aligned} -D_k = |D_k| \leq -L(\beta^{(k)})^\top(\beta^{(k+1)} - \beta^{(k)}) &\leq \Big(\sum_{j \in S_k} \nabla_j L(\beta_j^{(k)})\Big) \cdot \Big(\sum_{j \in S_k}(\beta_j^{(k+1)} - \beta_j^{(k)})\Big) \\ &= \|\nabla_k L(\beta_k^{(k)})\|_2 \cdot \|\beta^{(k+1)} - \beta^{(k)}\|_2 \end{aligned} \tag{8}$$

where $\nabla_k L(\beta_k^{(k)})$ denotes the gradient at the parameter sampled at step k. The last step follows from the fact that, as the optimizer of the previous step that did not include

k, $\beta^{(k)}$ satisfies $0 \in \nabla_j L(\beta^{(k)})$ for all $j \neq k$. By the triangle inequality and our initial assumption, the l_2 norm of the difference in β coefficients must be bounded such that

$$\|\beta^{(k+1)} - \beta^{(k)}\|_2 \leq \frac{R(\beta^{k+1}) + R(\beta^{(k)})}{C_1} \leq \frac{2C_1}{C_2}$$

Plugging this bound into Equation 8 completes the proof. □

Note that a key component of this theorem is that the l_2-norm of β must be bounded by the regularizer. Such a bound can be constructed in a myriad of ways. As a simple example, any l_q norm with $q < 2$ satisfies

$$\|\beta\|_2 \leq \|\beta\|_q$$

So, for instance, lasso and ElasticNet regularization terms both satisfy the constraint of Theorem 5.

This theorem tells us that if the subgradient $\nabla_k L$ shrinks with n at an appropriate rate, the update of parameter k will become asymptotically negligible. Using this idea, one can construct an approximation whose error becomes asymptotically negligible even after omitting a large number of parameters.

Corollary 1 (Asymptotic subset sampling). Consider a sampling scheme that never samples any parameters j satisfying $\|\nabla_j L(\beta_j^{(k)})\|_2 = o_P(r_n^2)$. Then, the early-stopping error satisfies

$$\|L(\hat{\beta}^{(m)}) - L(\hat{\beta})\| = o_P(r_n)$$

Proof of Corollary 1. The approximation process error can be decomposed into a sum of omitted terms

$$\|L(\hat{\beta}^{(m)}) - L(\hat{\beta})\| = \|\sum_{j=m}^{d} D_j\|$$

Since $D_j \asymp \|L(\beta_j^{(k)})\|_2 = o_P(r_n^2)$, the right hand side must be $o_P(r_n)$, completing the proof. □

This idea is most easily put into practice for models linear in some basis. In the next section, we use Theorem 5 to construct an asymptotically-equivalent approximation to a tensor product spline that can achieve an $o_P(n^{-1/4})$ rate of convergence.

5. Illustration: Tensor Product Splines. Let us see an example of how all the preceding theory can be applied in practice. Suppose one seeks to fit a conditional mean model $E(Y \mid X)$ over $X_1, \ldots X_n$ vectors of D covariates. We will consider the tensor product basis model

$$f_\beta(x) = \Phi(x)\beta = \sum_{i=1}^{n} \sum_{S \subset \{1,\ldots,D\}} \beta_{i,S} \prod_{j \in S} \mathbb{1}(x_j \leq X_{ij}) x_{ij}^q \tag{9}$$

Hence, f_β is a linear function over the collection of a qth-order smoothing spline basis functions $\phi_{i,S}(x)$, where i represents the knot point and S represents the interactions.

As discussed earlier, when $f^\star \in \mathscr{F}$ where $\mathscr{F}$ contains the set of all càdlàg functions with finite sectional variation, one can estimate $f^\star$ by solving the lasso-penalized least-squares problem

$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^d} L(\beta; O) = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \|Y - \Phi(X)\beta\|_2^2 + \|\beta\|_1 \ . \tag{10}$$

Even though this estimator is minimax-optimal over a large class of functions, it contains $n(2^p - 1)$ parameters; hence, solving the full loss minimization problem will typically be too computationally intensive for more than a few covariates, motivating an approximation. Note, however, that each basis function $\phi(x) = \mathbb{1}(x_j \leq X_{ij})x_{ij}^q$ is indexed by a knot i, a section S, and an exponent q. This means one can construct an approximation by sampling a subset of the full basis using the following algorithm:

Algorithm 1: RandomHAL

1. Set the number of basis functions to sample as $m \gtrsim n \log n$ (or $m \gtrsim n$ if sampling without replacement).
2. Construct an approximate basis $\Phi_m(x)$ by sampling m tuples (i, S, q) and constructing a basis function from each tuple.
3. Solve $\hat{\beta} = \arg\min \frac{1}{n}\sum_{i=1}^{n} \ell(\Phi(X)\beta) + \|\beta\|_1$ for the desired smooth loss ℓ.
4. The RandomHAL function estimate will be $f_{\hat{\beta}}(x) = \Phi(X)\hat{\beta}$.

As we argued in Remark 1 and Theorem 3, sampling $m \gtrsim n$ tuples uniformly without replacement or $m \gtrsim n\log(n)$ with replacement in Step 2 will guarantee that the corresponding RandomHAL estimator converges at an $o_P(n^{-1/4})$ rate. However, the finite-sample performance of this learning algorithm will in general still be quite bad for large D, as many basis functions will likely be very high-order interactions that evaluate as vectors of all zero, requiring “galactically large” n to overcome. Therefore, achieving good finite-sample performance will require a non-uniform sampling strategy that strategically excludes basis functions unlikely to reduce the loss.

Applying Theorem 5 to a linear basis, we can see that in order for a sampling scheme that omits certain parameters to perform better than uniform, we need any omitted parameters to satisfy

$$\|\nabla_j L(\beta_j^{(k)})\|_2 = \|T\Big(\frac{1}{n}\phi_j^\top(y - \Phi_{S_k}\beta^{(k)}), \lambda\Big)\|_2 = o_P(n^{-1/2}) \tag{11}$$

where T represents the soft-thresholding function $T(z, \lambda) = \operatorname{sign}(z)\max(|z| - \lambda, 0)$. Given a small amount of information about the form of the basis functions and the data generating process, several different types of basis functions may satisfy this condition and therefore be excluded from the sampling. We refer to an exclusion rule as a *heuristic*. Each heuristic lower bounds some portion of the left-hand side, providing a theoretically-justified strategy for improving finite-sample error. The idea is that if one has enough knowledge to deploy a heuristic at a given step, they should do so; otherwise, sample uniformly.

Example 1 (Choose basis functions correlated with residual). Equation (11) implies that $\|\nabla_j L(\beta_j^{(k)})\|_2 \asymp \phi_j^\top r/n$, where r represents the residual $r = y - \Phi\hat{\beta}^{(k)}$; that is, we ought to avoid sampling basis functions uncorrelated with the residuals produced after sampling other basis functions. Therefore, greedily sampling the “best” basis function—the one with the highest correlation at each step—will achieve good finite sample error.

Tree-based models that choose the “best” split minimizing some loss exploit the logic of Example 1. But evaluating parameter update is “best” can often be computationally intensive outside the context of trees. How do we target basis functions that have strong covariance with the residual? The next examples provide some specific strategies.

Example 2 (Asymptotically negligible basis functions). Suppose that the vector ϕ_j contains n/r_n^2 nonzero terms. This implies

$$\frac{1}{n} r^\top \phi_j(X) = o(1/r_n^2)$$

Hence, any basis excluding this term will only incur a maximum difference of $o(1/r_n^2)$ compared to the full basis. If the desired approximation rate to match is r_n, then excluding this term will still preserve the same rate as a uniform sampling scheme. Such logic justifies the common heuristic of fitting HAL with only the subset of basis functions with $\sqrt{n}$ nonzero entries (Hejazi, Coyle and van der Laan, 2020; Coyle et al., 2022). However, one must be warned not to exclude more than r_n^2 of these terms, or the rate will not be preserved, and the method is liable to suffer when predicting at the tails of the data distribution.

Example 3 (Drop high-order interactions). Suppose a tensor product main term satisfies $\phi_{ij} \in [0,1]$ – this occurs, for example, if X are normalized, or Φ encodes a zero-order spline. An interaction between main terms with indices given by $S \subset \{1, \ldots, d\}$ is

$$\frac{1}{n} \sum_{i=1}^{n} r_i \prod_{j \in S} \phi_{ij}$$

Now recall Lemma 1 which implies that

$$\prod_{j \in S} \phi_{ij} \leq \exp\Big(- \sum_{j \in S} \phi_{ij} \Big) \lesssim \exp(-|S|)$$

Hence, as the order of the interaction $|S|$ grows, this term will dissipate asymptotically; hence, one only need sample interactions on the log-order of the desired squared rate. For example, suppose one were to exclude any interactions such that $|S| \geq \frac{1}{2}\log(n)$. Then, the above simplifies to

$$\exp(-|S|) = n^{-1/2}$$

meaning that the gradient of any excluded term would be asymptotically negligible for an algorithm achieving an $o(n^{-1/2})$ rate. A simple rule of thumb: one roughly only needs interactions up to the *order of magnitude of the sample size plus one*. For example, if the dataset contains $n = 100$ observations, one should only need to include up to second-order interactions in the model (that is, products of up to three different main terms).

Example 4 (Drop high-order polynomial splines). Again suppose $\phi(x) \in [0,1]$; for example, if X are normalized or Φ represents a zero-order spline. Similarly to the previous example, the qth order polynomial of a main term ϕ_j will satisfy

$$\frac{1}{n} \sum_{i=1}^{n} r_i \phi_{ij}^q \lesssim \exp(-q)$$

so any polynomial terms of order $q \gtrsim \frac{1}{2}\log(n)$ will be asymptotically negligible and can be dropped.

Any of the above heuristics can be used to construct a RandomHAL sampling scheme that improves over the finite-sample error of uniform sampling while preserving the desirable asymptotic rate of HAL.

6. Sampling schemes that speed up computation. The random basis approximation schemes discussed so far preserve a desired asymptotic rate with a computationally tractable number of parameters, even in the presence of many covariates. Fitting such models typically relies on either gradient descent (for differentiable losses) or coordinate descent (for non-smooth losses, such as the lasso). The time complexity of computing a gradient or updating a set of coordinates for a model fit over n units with m parameters is generally $O(nm)$. For example, a rate-preserving approximation (sampling without replacement) of a tensor product spline such as HAL will have a time complexity of $O(n^2)$. This is still markedly slower than many existing models; for example, tree-based models only require $O(n \log n)$ complexity. In this section, we describe how randomness can be exploited to further reduce the computational complexity of fitting random basis models to be comparable to the fastest commonly-used tree-based models.

The idea will to sample basis functions in blocks that admit fast matrix computation. Let Φ denote the random basis, and $\odot$ the element-wise (Hadamard) matrix product. Suppose this basis admits a decomposition

$$\Phi = \left[U_1 \odot F_1 \ldots U_{T_n} \odot F_{T_n}\right]$$

where U_t denotes a *low-rank* matrix and F_t a matrix with a fast $O(n)$ matrix multiplication. Then, the matrix multiplications $\Phi\beta$ and Φv used in fitting the model would be $O(n \log n)$, meaning that computing a random basis approximation would have the same time complexity as commonly used tree-based models.

How can organize such a decomposition for a tensor product spline? Generally, the strategy will be to let F_t denote the "indicator" part of the spline, and U_t the functional form of the spline. For example, let S denote the section of the spline, and q the order of the polynomial. Then,

$$U_t(x) = \Big[\prod_{j\in S} x_j^q - \prod_{j\in S} X_{ij}^q\Big]_{i=1}^{n_t} \tag{12}$$

clearly admits a low-rank representation, as it can be represented by the outer product

$$\left[\prod_{j\in S} x_j^q \; -1\right] \begin{bmatrix} 1 \\ \prod_{j\in S} X_{ij}^q \end{bmatrix}_{i=1}^{n_t}$$

With the functional part of the spline taken care of via L_t, let us consider a matrix of indicators F_t. How do we ensure such a matrix admits a fast matrix-vector multiplication? We will consider a *nested* basis. Suppose we can partition the sample space into nested regions $\mathcal{R}_1 \subset \mathcal{R}_2 \subset \ldots \subset \mathcal{R}_{n_t}$. Then, construct a basis

$$F_t(x) = \left[\mathbb{I}(x \in \mathcal{R}_1) \ldots \mathbb{I}(x \in \mathcal{R}_{n_t})\right]$$

This matrix will admit a fast matrix-vector multiplication due to the recursive structure of its inner product:

$$\langle I(x \in R_i), v\rangle = \langle I(x \in R_{i-1}), v\rangle + \langle I(x \in R_i - R_{i-1}), v\rangle$$

Hence, if v is of length n, the inner product can be computed in $O(n)$ by simply summing the elements of v sequentially and recording every sum that exhausts the elements of a given region. This is how 1-dimensional smoothing splines can be fit quickly—after sorting the data, the spline indicators form a triangular matrix. However, the same strategy works for high dimensions, provided we design a sampling scheme to *only* sample cut points that are nested within one another.

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Fig 2: Example of a 4×4 "nested" matrix.

What about U_t? A convenient property of the Hadamard product $\odot$ is that it preserves fast matrix multiplication with low-rank matrices. For a low-rank matrix representable as $U = \sum_{i=1}^r x_i y_i^\top$ for some vectors $(x_i, y_i)_{i=1}^r$ and an arbitrary matrix F,

$$(U \odot F)v = \sum_{i=1}^r D_{x_i} F D_{y_i} v$$

where D_x denotes a matrix with diagonal vector x and zero everywhere else. This representation allows us to exploit Hadamard products of low-rank and "nested" matrices. Let block t of the partition of Φ contain basis functions of interaction S and polynomial order q. Then, letting $D^{(q,S)}$ denote a diagonal matrix such that $D_{ii}^{(q,S)} = \prod_{j \in S} X_{ij}^q$,

$$(U_t \odot F_t)v = D_{ii}^{(q,S)} F_t(X)v - F_t(X) D_{ii}^{(q,S)} v \ ,$$

which can be computed in $O(n)$ time by definition of a diagonal matrix. Hence, if $T_n = O(\log n)$, computing the entire $\Phi(X)v$ will be $O(n \log n)$ as desired. In fact, it will be $O(dn \log n)$, since the bottleneck will be sorting the data in order to construct partitions—the same as in a tree-based model. This construction thereby extends the computational convenience of a tree to general tensor product models.

A note on centering: When fitting a lasso model, it is customary to center and scale each column of the basis matrix. When Φ is not stored in memory, the scaling must be done during the optimization process by precomputing the mean and variance of each column. Computing the mean of each column can be easily accomplished using the fast matrix multiplication $\bar{\Phi} = \frac{1}{n}\Phi^\top 1$. Computing the variance, however, is slightly trickier. We exploit the property $\mathrm{Var}(X) = E(X^2) - E(X)^2$ column-wise across the matrix. The sum of squares can be expressed $\frac{1}{n}(\Phi \odot \Phi)^\top 1$. The nested portion $F_t \odot F_t = F_t$ since its entries are all 1 or 0. The low-rank portion, because it can be expressed $U_t = x1^\top - 1y^\top$, satisfies $U_t \odot U_t = x^2 1^\top + 1y^{2\top} - 2xy^\top = U_t'$, another low-rank matrix (powers are applied element-wise). Then, each block of $\Phi \odot \Phi$ satisfies $U_t \odot F_t \odot U_t \odot F_t = U_t' \odot F_t$, which admits a fast matrix multiplication by the same principles discussed previously. Letting Φ' be the basis of blocks $U_t' \odot F_t$, the variance of each column of Φ can be computed in $O(n)$ by the computation $\hat{\sigma}^2 = \frac{1}{n}\Big((\Phi')^\top 1 - \frac{1}{n}(\Phi^\top 1)^2\Big)$, with the square in the second term applied element-wise across the output vector.

7. Numerical experiments.

Confirmatory simulation experiments.

Include a more standard / simple example other than HAL. Add links to notebooks similar to the Prediction Powered Inference paper.

8. Discussion.

A call for non-asymptotic analysis in causal inference, rather than a total dependence on rates. Generalization error can be bounded by virtue of being in a Donsker class.

An example function with a loss solely based on X is the Riesz representer, the function f satisfying the balancing equation $\mathsf{E}[g(h; X)] = \mathsf{E}[f(X_i)h(X_i)]$ for all h. This minimizes $\mathsf{E}[f(X_i)^2 - 2g(f; X_i)]$ and has gained popularity in the fields of causal inference and missing data problems Chernozhukov et al. (2021); Williams, Hines and Rudolph (2025)

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REFERENCES

ATHEY, S., TIBSHIRANI, J. and WAGER, S. (2019). Generalized random forests. *The Annals of Statistics* **47**. https://doi.org/10.1214/18-aos1709

BENKESER, D. and VAN DER LAAN, M. (2016). The Highly Adaptive Lasso Estimator. In *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)* 689–696. IEEE. https://doi.org/10.1109/dsaa.2016.93

BIAU, G. (2012). Analysis of a random forests model. *The Journal of Machine Learning Research* **13** 1063–1095.

BIAU, G., DEVROYE, L. and LUGOSI, G. (2008). Consistency of random forests and other averaging classifiers. *Journal of Machine Learning Research* **9**.

BIBAUT, A. F. and VAN DER LAAN, M. J. (2019). Fast rates for empirical risk minimization over càdlàg functions with bounded sectional variation norm. *arXiv*. https://doi.org/10.48550/ARXIV.1907.09244

BOAS, R. P. and WRENCH, J. W. (1971). Partial Sums of the Harmonic Series. *The American Mathematical Monthly* **78** 864–870. https://doi.org/10.1080/00029890.1971.11992881

BOSER, B. E., GUYON, I. M. and VAPNIK, V. N. (1992). A training algorithm for optimal margin classifiers. In *Proceedings of the fifth annual workshop on Computational learning theory. COLT92* 144–152. ACM. https://doi.org/10.1145/130385.130401

BREIMAN, L. (2001). Random Forests. *Machine Learning* **45** 5–32. https://doi.org/10.1023/a:1010933404324

BREIMAN, L., FRIEDMAN, J. H., OLSHEN, R. A. and STONE, C. J. (2017). *Classification And Regression Trees*. Routledge. https://doi.org/10.1201/9781315139470

BÜHLMANN, P. and YU, B. (2003). Boosting With theL2Loss: Regression and Classification. *Journal of the American Statistical Association* **98** 324–339. https://doi.org/10.1198/016214503000125

CHEN, T., HE, T., BENESTY, M., KHOTILOVICH, V., TANG, Y., CHO, H., CHEN, K., MITCHELL, R., CANO, I., ZHOU, T., LI, M., XIE, J., LIN, M., GENG, Y., LI, Y., YUAN, J. and CORTES, D. (2025). xgboost: Extreme Gradient Boosting R package version 3.0.5.1.

CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., DUFLO, E., HANSEN, C., NEWEY, W. and ROBINS, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21** C1–C68. https://doi.org/10.1111/ectj.12097

CHERNOZHUKOV, V., NEWEY, W. K., QUINTAS-MARTINEZ, V. and SYRGKANIS, V. (2021). Automatic Debiased Machine Learning via Riesz Regression. https://doi.org/10.48550/ARXIV.2104.14737

CHIPMAN, H. A., GEORGE, E. I. and MCCULLOCH, R. E. (2010). BART: Bayesian additive regression trees. *The Annals of Applied Statistics* **4**. https://doi.org/10.1214/09-aoas285

CORTES, C. and VAPNIK, V. (1995). Support-vector networks. *Machine Learning* **20** 273–297. https://doi.org/10.1007/bf00994018

COYLE, J. R., HEJAZI, N. S., PHILLIPS, R. V., VAN DER LAAN, L. and VAN DER LAAN, M. J. (2022). hal9001: The scalable highly adaptive lasso. R package version 0.4.2. https://doi.org/10.5281/zenodo.3558313

FANG, B., GUNTUBOYINA, A. and SEN, B. (2021). Multivariate extensions of isotonic regression and total variation denoising via entire monotonicity and Hardy–Krause variation. *The Annals of Statistics* **49**. https://doi.org/10.1214/20-aos1977

FRIEDMAN, J. H. (1991). Multivariate Adaptive Regression Splines. *The Annals of Statistics* **19**. https://doi.org/10.1214/aos/1176347963

GAO, W., XU, F. and ZHOU, Z.-H. (2022). Towards convergence rate analysis of random forests for classification. *Artificial Intelligence* **313** 103788. https://doi.org/10.1016/j.artint.2022.103788

HEJAZI, N. S., COYLE, J. R. and VAN DER LAAN, M. J. (2020). hal9001: Scalable highly adaptive lasso regression in R. *Journal of Open Source Software*. https://doi.org/10.21105/joss.02526

KIMELDORF, G. and WAHBA, G. (1971a). Some results on Tchebycheffian spline functions. *Journal of mathematical analysis and applications* **33** 82–95.

Kimeldorf, G. and Wahba, G. (1971b). Some results on Tchebycheffian spline functions. *Journal of Mathematical Analysis and Applications* **33** 82–95. https://doi.org/10.1016/0022-247x(71)90184-3

Klaassen, C. A. J. (1987). Consistent estimation of the influence function of locally asymptotically linear estimators. *The Annals of Statistics* 1548–1562.

Koshevnik, Y. A. and Levit, B. Y. (1977). On a non-parametric analogue of the information matrix. *Theory of Probability & Its Applications* **21** 738–753.

Kugarajah, T. and Zhang, Q. (1995). Multidimensional wavelet frames. *IEEE Transactions on Neural Networks* **6** 1552–1556. https://doi.org/10.1109/72.471353

Lin, Y. (2000). Tensor product space ANOVA models. *The Annals of Statistics* **28**. https://doi.org/10.1214/aos/1015951996

Mercer, J. (1909). Functions of positive and negative type, and their connection the theory of integral equations. *Philosophical Transactions of the Royal Society of London* 415-446. https://doi.org/10.1098/rsta.1909.0016

Peng, W., Coleman, T. and Mentch, L. (2022). Rates of convergence for random forests via generalized U-statistics. *Electronic Journal of Statistics* **16**. https://doi.org/10.1214/21-ejs1958

Pfanzagl, J. and Wefelmeyer, W. (1985). Contributions to a general asymptotic statistical theory. *Statistics & Risk Modeling* **3** 379–388.

Qiu, T. (2023). A basis adaptive sampling approach for the optimization of highly adaptive lasso, PhD thesis, University of California, Berkeley.

Qiu, H., Luedtke, A. and Carone, M. (2021). Universal sieve-based strategies for efficient estimation using machine learning tools. *Bernoulli* **27** 2300–2336.

Rahimi, A. and Recht, B. (2007). Random features for large-scale kernel machines. *Advances in neural information processing systems* **20**.

Rahimi, A. and Recht, B. (2008). Uniform approximation of functions with random bases. In *2008 46th Annual Allerton Conference on Communication, Control, and Computing* 555–561. IEEE. https://doi.org/10.1109/allerton.2008.4797607

Raskutti, G., Wainwright, M. J. and Yu, B. (2014). Early stopping and non-parametric regression: an optimal data-dependent stopping rule. *The Journal of Machine Learning Research* **15** 335–366.

Rendle, S. (2010). Factorization machines. In *2010 IEEE International conference on data mining* 995–1000. IEEE.

Ročková, V. and van der Pas, S. (2020). Posterior concentration for Bayesian regression trees and forests. *The Annals of Statistics* **48**. https://doi.org/10.1214/19-aos1879

Schuler, A., Hagemeister, A. and van der Laan, M. (2024). Highly Adaptive Ridge. https://doi.org/10.48550/ARXIV.2410.02680

Schuler, A., Li, Y. and van der Laan, M. (2023). Lassoed Tree Boosting. *arXiv*. https://doi.org/10.48550/ARXIV.2205.10697

Selesnick, I. W., Baraniuk, R. G. and Kingsbury, N. C. (2005). The dual-tree complex wavelet transform. *IEEE Signal Processing Magazine* **22** 123–151. https://doi.org/10.1109/msp.2005.1550194

Stone, C. J. (1994). The Use of Polynomial Splines and Their Tensor Products in Multivariate Function Estimation. *The Annals of Statistics* **22**. https://doi.org/10.1214/aos/1176325361

Syrgkanis, V. and Zampetakis, M. (2020). Estimation and inference with trees and forests in high dimensions. In *Conference on learning theory* 3453–3454. PMLR.

van der Laan, M. J. (2015). A generally efficient targeted minimum loss based estimator Technical Report No. 343, University of California, Berkeley. UC Berkeley Division of Biostatistics Working Paper Series.

van der Laan, M. (2017). A Generally Efficient Targeted Minimum Loss Based Estimator based on the Highly Adaptive Lasso. *The International Journal of Biostatistics* **13**. https://doi.org/10.1515/ijb-2015-0097

van der Laan, M. (2023). Higher Order Spline Highly Adaptive Lasso Estimators of Functional Parameters: Pointwise Asymptotic Normality and Uniform Convergence Rates. *arXiv*. https://doi.org/10.48550/ARXIV.2301.13354

van der Laan, M. J. and Robins, J. M. (2003). *Unified Methods for Censored Longitudinal Data and Causality*. Springer. https://doi.org/10.1007/978-0-387-21700-0

van der Laan, M. J. and Rose, S. (2011). *Targeted Learning: Causal Inference for Observational and Experimental Data* **4**. Springer.

van der Laan, M. J. and Rubin, D. (2006). Targeted maximum likelihood learning. *The International Journal of Biostatistics* **2**.

WAGER, S. and ATHEY, S. (2018). Estimation and Inference of Heterogeneous Treatment Effects using Random Forests. *Journal of the American Statistical Association* **113** 1228–1242. https://doi.org/10.1080/01621459.2017.1319839

WAINWRIGHT, M. J. (2019). *High-dimensional statistics: A non-asymptotic viewpoint* **48**. Cambridge university press.

WANG, M., SCHULER, A., VAN DER LAAN, M. and MEIXIDE, C. G. (2026). Highly Adaptive Principal Component Regression. https://doi.org/10.48550/ARXIV.2602.10613

WILLIAMS, N. T., HINES, O. J. and RUDOLPH, K. E. (2025). Riesz representers for the rest of us. https://doi.org/10.48550/ARXIV.2507.19413

WILLIAMS, C. K. I. and RASMUSSEN, C. E. (1995). Gaussian processes for regression. In *Proceedings of the 9th International Conference on Neural Information Processing Systems. NIPS'95* 514–520. MIT Press, Cambridge, MA, USA.

YANG, Y. and BARRON, A. (1999). Information-theoretic determination of minimax rates of convergence. *The Annals of Statistics* **27**. https://doi.org/10.1214/aos/1017939142

ZHANG, T. and SIMON, N. (2023). Regression in tensor product spaces by the method of sieves. *Electronic Journal of Statistics* **17**. https://doi.org/10.1214/23-ejs2188